Hedging or Harvesting?

Classical versus Learned Option Hedging under Transaction Costs

Replications of Sepp (2013) and Buehler et al. (2019), with a head-to-head comparison

DSC4107 Final Project

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The problem

- Sell an at-the-money one-month call, hedge it with the underlying.
- Trades happen at discrete times, and each one costs a fixed fraction of notional.
- More trading tracks the delta but pays more cost; less trading saves cost but leaves variance.

Two papers attack the same trade-off from opposite ends:

Sepp (2013)
Closed-form P&L statistics for delta hedging. Maximises the Sharpe ratio over rebalancing frequency, then converts it to price and delta bands.

Buehler et al. (2019)
A neural network chooses the hedge from market state, trained on a convex risk measure of terminal P&L.

Neither paper runs the two against each other under controlled conditions. That comparison is our contribution.

Research questions

- Can a Monte Carlo simulator reproduce Sepp's closed-form statistics, tables and figures?
- Can we reproduce the main claim of Deep Hedging, that a trained network beats Black-Scholes delta hedging under transaction costs in a Heston market?
- On one testbed, which approach gives the better risk-adjusted P&L, and does the ranking change with the market model and the size of the transaction cost?

The answer to RQ3 turns out to depend on what "better" means: Sharpe or tail risk.

Methods

Testbed

- Four market models: GBM, jump-diffusion, Heston, Heston with jumps
- Sepp's parameters, so replication lines up
- One ATM one-month call, one proportional cost, one shared P&L rollout
- Cost charged on opening and closing trades too

Comparison design

- Classical baseline re-tuned in every cell, so a stale setting cannot manufacture a win
- Learned network: state is moneyness, time, current position; CVaR loss
- Disjoint train, validation, tuning, search and holdout seed pools
- Every number averages three training seeds; per-seed minimum and win rate reported

Correctness anchor. With zero cost the learned policy must recover the Black-Scholes delta. If it does not, no learned result is trustworthy.

Replication 1: Sepp (2013)

- Closed forms for mean P&L `u`, cost `c`, local variance `p`, variance floor `f`
- Largest relative error in `u`: 3.9%
- GBM and SVJ case I match to 0.0001
- Monte Carlo cross-check (60,000 paths) agrees on mean and volatility

The jump-diffusion fix

The JD branch had dropped a factor `q = sqrt(pi^3)/4` that the other branches carry. Restoring it moves `p` from 24% below the paper to 5.6% above and brings the analytic JD Sharpe onto the simulated one.

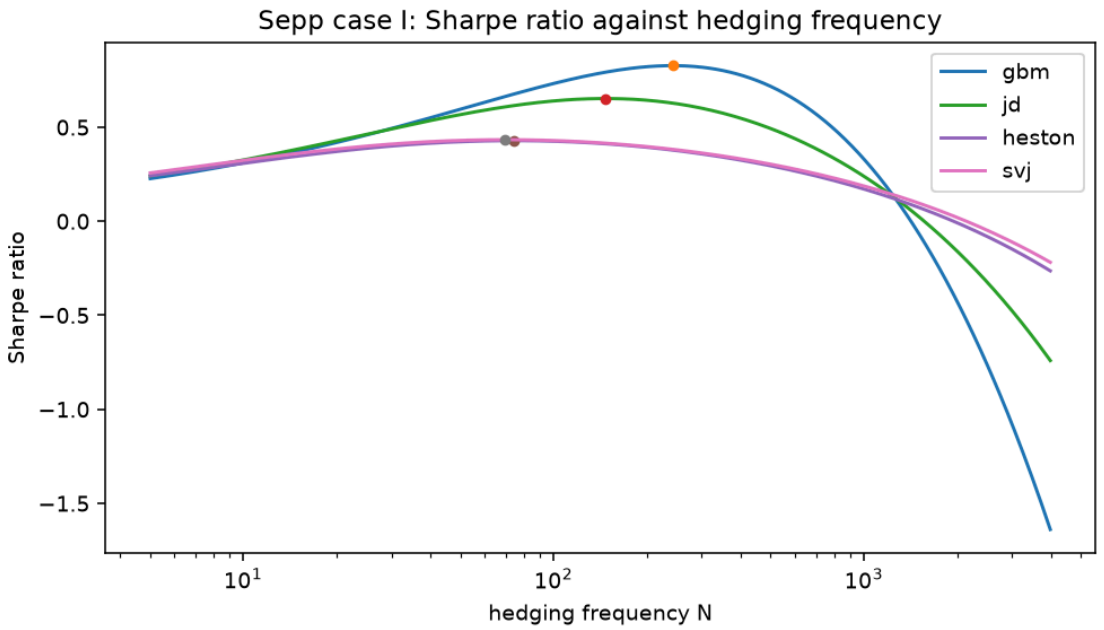


Figure 1: Sharpe against hedging frequency, case I.

Replication 2: Deep Hedging

- Frictionless check: correlation 0.9907, slope 1.148, RMSE 0.0581
- The learned policy rediscovers the delta, which validates the implementation
- In Heston, it matches the delta hedge at zero cost and pulls ahead as cost rises
- This reproduces the paper's qualitative claim

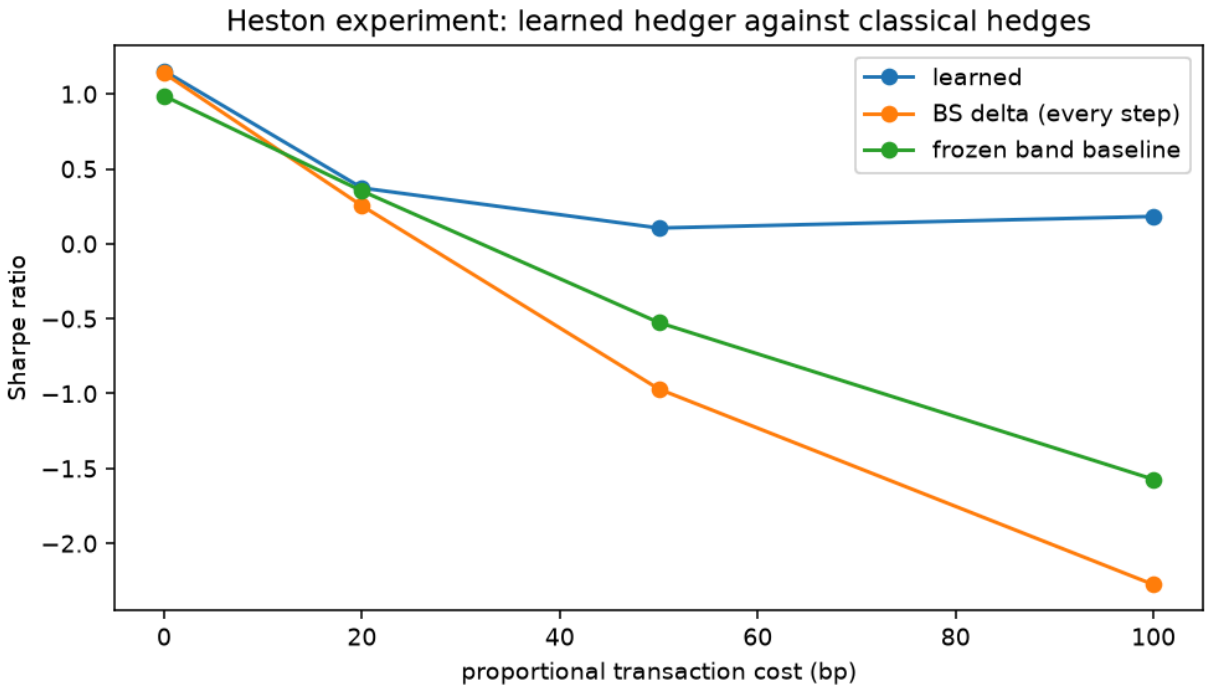


Figure 2: Heston Sharpe against cost.

The head-to-head comparison

- **Near zero cost:** the two approaches tie, as the frictionless check predicts
- **At 20 bp:** flat, edges from −0.09 to +0.02; the jump models favour classical
- **Above 50 bp:** a large Sharpe edge, but the P&L standard deviation triples

The high-cost edge is the volatility risk premium being harvested, not better hedging.

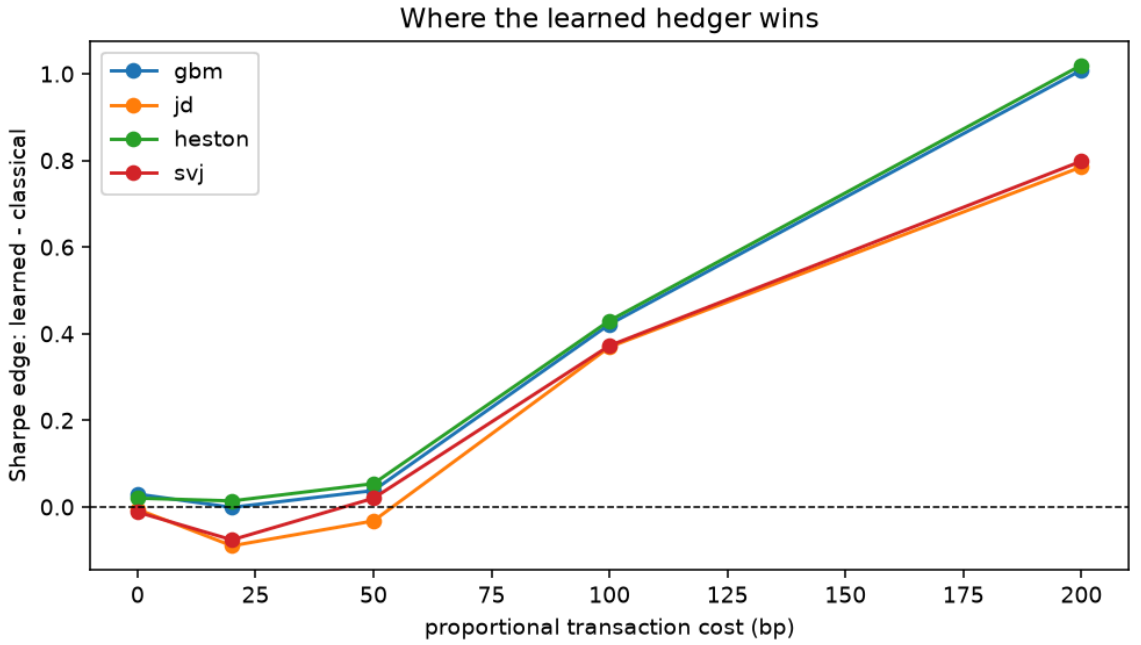


Figure 3: Learned minus classical Sharpe across four models and five costs.

Is the moderate-cost edge real?

- Locked holdout, five unseen seeds: mean edge +0.0339, minimum +0.0293, win rate 1.00
- Bootstrap intervals on the validation pool exclude zero
- So the small Heston 20 bp edge is not seed or path noise

But it is not free.

- The learned P&L has the same spread as the band
- Skew −0.70 against −0.01
- CVaR at 95% is 24% worse

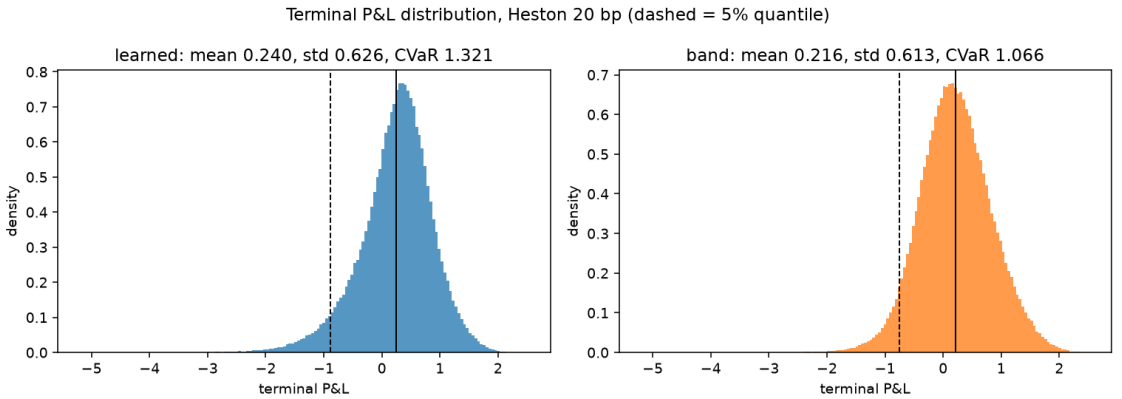


Figure 4: Terminal P&L on the frozen benchmark.

Hedging or harvesting: the mechanism

- The CVaR tail fraction `alpha` is the dial. Large `alpha` collapses turnover and triples the P&L spread: the network stops hedging and collects the premium.
- Removing the current position from the state costs Sharpe and raises turnover, so position feedback produces the band.
- A participation dial `lambda` in the classical family tunes "how much delta" and "how often".
- At 0-20 bp the tuned dial is near the textbook delta and the network stays 0.02 to 0.05 above it.
- Above 20 bp the dial goes to zero, and the network is below do-nothing.

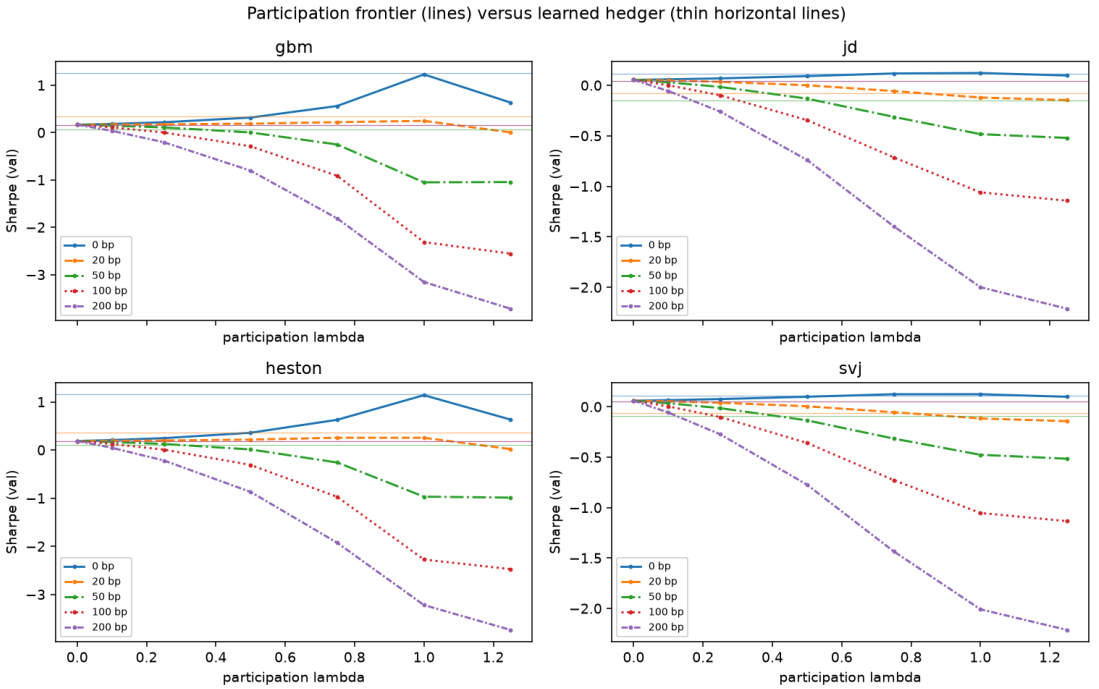


Figure 5: Participation frontier.

Conclusion

- The extra freedom of a neural hedger does not buy a better hedge in the benchmark regime.
- At 20 bp the network ties a well-tuned classical delta band, and the surviving holdout edge comes with a worse tail.
- Where it posts large gains, above roughly 50 bp, it has stopped hedging, and a classical family with the same freedom does that better.
- The replications hold up: Sepp's coefficients match Table 2, the closed forms agree with Monte Carlo, and the frictionless network recovers Black-Scholes.

Report any learned hedging edge next to its P&L spread. Sharpe alone hides the change of risk profile.

Reproduce it

github.com/hluaguo/dsc4107

```
git clone https://github.com/hluaguo/dsc4107.git
cd dsc4107
uv sync
uv run pytest -q                          # 24 checks
uv run python scripts/sepp_replication.py
uv run python scripts/deep_hedging_replication.py
uv run python scripts/comparison_study.py --fixed-hyper
uv run python scripts/holdout_eval.py
```

Full tables, source code and raw results are in the written report appendices and the `submission/` package.